

Material Test Report / Certificate GAT-24-05521

Gulf Alloy Tubulars LLC | 1400 Sheldon Rd, Channelview, TX 77530

Certificate No. / Date	GAT-24-05521 / 2024-06-03
Heat / Batch No.	K73209
Customer	American Completion Tools, Inc.
Purchase Order	ACT PO-86912
Product	Seamless mechanical tube, 6.10 in. OD x 1.20 in. wall, Q&T;
Specification(s)	API 5CT L80 Type 13Cr (UNS S42000 mod.); NACE MR0175/ISO 15156-3
Quantity	8 tubes / 9,850 lb

Chemical Composition (wt %)

Element	Value	Element	Value
C	0.19	Cr	12.85
Mn	0.46	Mo	0.02
P	0.017	Ni	0.21
S	0.004	Cu	0.11
Si	0.31	Fe	Bal.

Mechanical / Physical Properties

Property	Result	Requirement
Yield strength (0.2% offset)	88,400 psi	80,000 - 95,000 psi
Tensile strength	102,600 psi	>= 95,000 psi
Elongation in 2 in.	21 %	>= 15 %
Hardness (max of 4 quadrants)	22.5 HRC	<= 23 HRC
Charpy V-notch @ 32 degF (avg of 3)	58 ft-lb	>= 20 ft-lb

Thermal Treatment

Austenitized 1,780 degF, air quench; tempered 1,250 degF; straightened; stress relieved 1,000 degF

NDE / Additional Testing

Full-length EMI + UT wall per API 5CT SR2: accepted

Sour Service (NACE MR0175 / ISO 15156) Statement

Certified compliant with NACE MR0175/ISO 15156-3 for sour service (hardness <= 23 HRC).

Remarks

Certificate per EN 10204 Type 3.1. Used by ACT for: PKR-5500 mandrels (heat traceable to BOM item ACT-PKR-5500-02), LH-7625 hanger bodies and PBR (see QIR-2026-0031), 13Cr trim options.

Certification

We hereby certify that the material described above was manufactured, sampled, tested and inspected in accordance with the referenced specifications and was found to meet those requirements. This certificate is issued in accordance with EN 10204 Type 3.1.

Role	Name	Signature	Date
Authorized Signatory	H. Delacroix, QA Manager, Gulf Alloy Tubulars	/signed electronically/	2024-06-03
ACT Receiving QA Review	S. Nguyen	/signed electronically/	2024-06-03